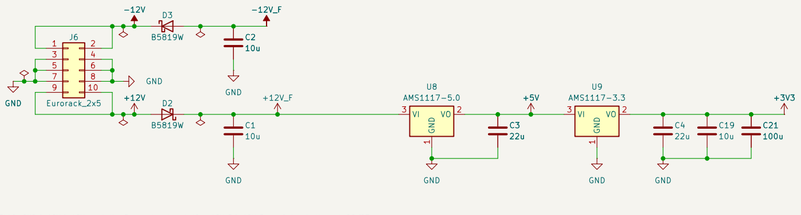
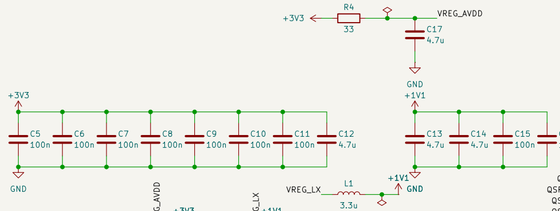
BIG MODULAR 16BIT — S1 schematic (Bread Modular 16bit for Eurorack, 6HP, 4 layers). Circuit intent: docs/research.md; placement: docs/geometry.json

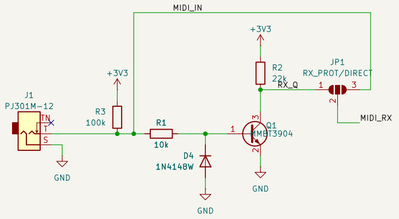
POWER: +12V -> D2 -> +12V_F (U5 V+, U8 AMS1117-5.0 -> +5V -> U9 AMS1117-3.3 -> +3V3); -12V -> D3 -> -12V_F (U5 V-)



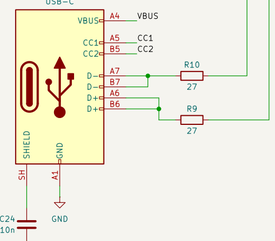
PWR_FLAG: GND/+12V/-12V mark the Eurorack entry at J6; +12V_F/-12V_F are fed only through D2/D3; +1V1 (core regulator via L1) and VREG_AVDD (via R4) are fed through passive parts.

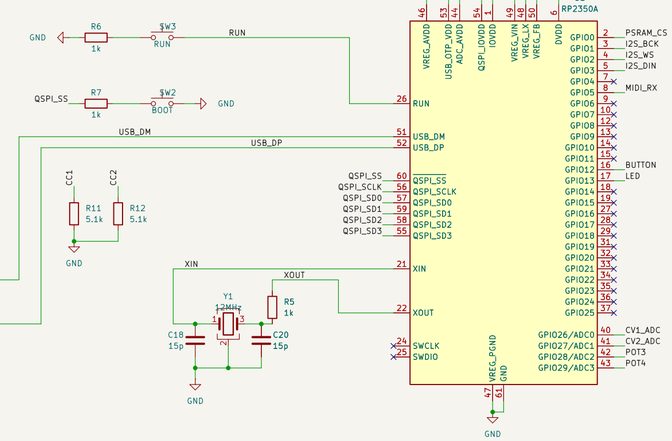


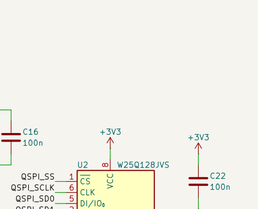
MIDI IN (TS): MIDI V2 / 8bit NPN front end at 3.3 V, no opto



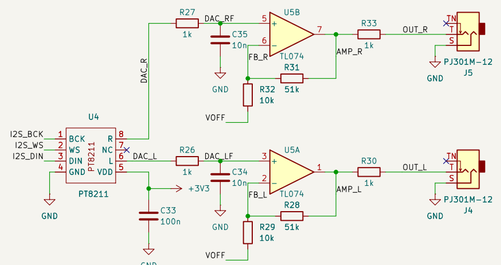
JP1 RX SOURCE (back, shipped OPEN, bridge one pair): 1–2 = PROTECTED (Q1 drives GPIO5, firmware inverts the input); 2–3 = DIRECT (unprotected, stock BM 16bit firmware). Never both.

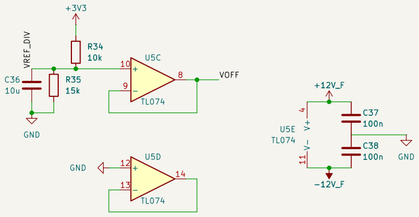




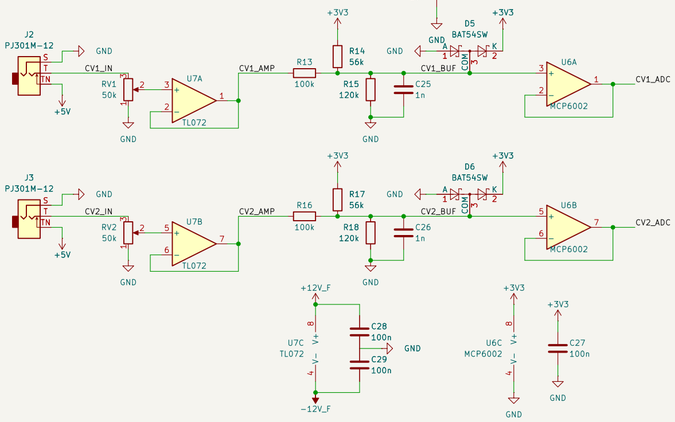


DAC + OUTPUTS: PT8211 (I2S GPIO1–3) -> 1k/10n -> U5A/U5B gain 6.1 with −5.1 x VOFF (1.98 V): full scale 1.65 V +/− 0.825 V -> 0 V +/− 5.0 V, DC coupled, 1k series to J4 (L) / J5 (R).

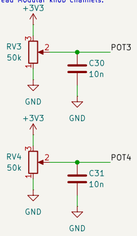




CV IN 1/2: tip normalled to +5V -> RV1/RV2 50k attenuator -> U7 TL072 follower -> 100k / 56k to +3V3 / 120k to GND; BAT54SW clamp, 1n -> U6 MCP6002 follower -> ADC0/ADC1. Unpatched: RV = 0..+5 V knob. Patched: RV scales the CV, −5.9..+6.0 V -> 0..3.3 V.

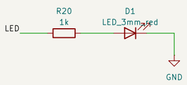


KNOBS 3/4: 50k dividers on +3V3 -> ADC2 (GPIO28) / ADC3 (GPIO29), the Bread Modular knob channels.



MODE: SW1 (6 x 6 mm THT tact) on GPIO12, red LED on GPIO13 through 1k (Bread Modular pins). GPIO13 is PWM6B: the duty cycle sets the LED brightness.





Circuit intent: docs/research.md; placement: docs/geometry.json
SCHEMATIC S1 — Bread Modular 16bit (RP2350) for Eurorack 6HP: MIDI in, 2 CV in, 4 pots, red mode

Big Modular

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Title: Big Modular — 16bit

Size: A2	Date:	**Rev: S1**
KiCad E.D.A. 10.0.6		Id: 1/1